John Wesley Agojo

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Education

University of Massachusetts Lowell — Lowell, MA
Bachelor of Science in Computer Science — General — *Expected: December 2027*
Minor: Mathematics — 3.8 GPA

- **Coursework:** Data Structures and Algorithms, Probability and Statistics, Discrete Mathematics 2, Calculus 3, Object-Oriented Programming (C/C++), Cloud Computing, Graphical User Interface, Computer Architecture

Professional Experience

Savvas — Boston, MA
Software Engineer Intern — *June 2026 - Aug 2026*

- Engineered a company-wide documentation tool adopted across **50+ repositories** that monitors pull requests, summarizes code changes via **AWS Bedrock (Claude Haiku)**, and posts ranked **Confluence** update suggestions directly on PRs.
- Developed a **Python** pipeline using the **GitHub REST API** to fetch PR diffs, applying a category-aware compression strategy (50,000-token budget) to safely process massive PRs prior to **Bedrock** summarization.
- Implemented hybrid vector/lexical retrieval across a **Bedrock Knowledge Base** with **Cohere native reranking**, scoring Confluence pages via custom heuristics to surface the most relevant docs per PR.
- Built a scheduled **GitHub Actions** pipeline to export Confluence pages to Markdown, push updates to **S3**, and trigger Bedrock KB ingestion, packaged as a **Composite Action** for seamless adoption.

IBM — Lowell, MA
Software Developer Intern — *January 2026 - May 2026*

- Built the **FVT Diagnostic Tool**, an AI-powered triage system using a **3-layer LangGraph** state machine to classify IBM MAS FVT failures as test script, environment, or product issues via **IBM Bob** with **RAG**-augmented retrieval, auto-filing and deduplicating defects through the **Jira** REST API to reduce manual triage from hours to minutes.
- Engineered the tool's multi-phase eligibility gate using stagnation detection, version change analysis, and Ansible readiness checks to filter pipeline noise, propagating valid builds through an **Apache Kafka** event-driven architecture with partition-keyed ordering and manual offset commits.
- Queried FVT execution data from a pre-existing **MongoDB** instance and architected a new database collection for failure logging, then built a **Flask** backend with **50+** REST API endpoints and dynamic template rendering for pipeline monitoring, flaky test analysis, and defect tracking.

Vierra Digital — Cambridge, MA
Software Engineer Intern — *June 2025 - Sep 2025*

- Worked on customer-facing pages with **Next.js(React, Tailwind, Typescript).**
- Engineered an **AI-driven** ad automation agent that analyzes client websites, extracts marketing assets using **LLM** analysis, and generates platform-specific copy and creatives to publish campaigns.
- Created and optimized a **PostgreSQL** data layer to store website analysis, creative assets, campaign configurations, and performance metrics, which offers real-time insights into each client's **AI** agent.

Projects

AudioByte | *AWS (S3, DynamoDB, Lambda, Cognito), React.js, JavaScript, GraphQL, Tailwind CSS, Python CDK*

- Developed a **full-stack cloud-native** music streaming platform with secure authentication, real-time audio playback, and personalized library management using **React.js** and **AWS serverless** infrastructure.
- Engineered scalable backend using **AWS CDK** to provision **S3** for audio storage, **DynamoDB** for metadata, **Lambda** for serverless compute, and implemented a **GraphQL API** for efficient data queries.

Leadership Experience

UML The Filipino Club — Lowell, MA
President — *Sep 2025 - Present*

- Directed and mentored an executive board of 10 officers, delegating responsibilities and overseeing project timelines for all cultural and outreach initiatives.
- Grew member engagement through fostering an inclusive club culture and implementing effective communication channels for project management and event planning.

Skills, Technical Tools & Activities

Activities: Amazon IET-MSI program participant
Languages: C++, C, Python, TypeScript, HTML/CSS, JavaScript, Java, SQL, GraphQL
Web Technologies: React.js, Next.js, Flask, Tailwind CSS, JWT
AI & ML: Retrieval-Augmented Generation (RAG), Amazon Bedrock, Large Language Models (LLMs), Generative AI, OpenCV, YOLOv8
Technologies & Architecture: GitHub Actions (CI/CD), AWS CDK, Serverless Architecture, REST API, Git
Developer Tools: Github, Linux, Azure, AWS, OpenSearch, Confluence, JIRA, PostgreSQL, MongoDB, Scrum, Agile, FVT